

AI-Powered Job Recruitment & Candidate Management System

SmartBridge Internship Project

Program: SmartBridge Internship *(Implemented through SkillWallet & Salesforce)*

Project Team

Project Title:

AI-Powered Recruitment & Candidate Management System

Team ID: SWTID-2026-4435

Team Members

S. No	Name	Role
1	Lakshmi Aiswarya Akella	Team Lead
2	Radhika Satya Malleswari Mandapati	Member

Acknowledgement

We sincerely express our gratitude to the SmartBridge Internship Program, Salesforce, SkillWallet, and the faculty members of Ideal Institute of Technology for their continuous guidance and support throughout this project.

We would also like to thank our mentors for providing valuable knowledge on Salesforce CRM, Flow Automation, Prompt Builder, Agentforce, and other Salesforce technologies, which helped us successfully develop this AI-Powered Recruitment & Candidate Management System.

Finally, we thank everyone who encouraged and supported us during the design, development, testing, and successful completion of this project.

1. Project Overview

Introduction

The **AI-Powered Job Recruitment & Candidate Management System** is a centralized Salesforce application developed to modernize and automate the recruitment lifecycle using Artificial Intelligence, Salesforce Flow Automation, and Agentforce.

Traditional recruitment processes often require recruiters to manually review candidate applications, evaluate resumes, schedule interviews, approve salary packages, and monitor hiring progress. These activities consume significant time and may lead to inconsistent hiring decisions.

This project addresses these challenges by integrating Salesforce's automation capabilities with AI-powered decision support. The system intelligently prioritizes candidates, recommends suitable salary ranges, identifies hiring risks, and automates approval workflows, enabling recruiters and hiring managers to make faster and more informed decisions.

The application provides a centralized platform where recruiters can manage candidate information, monitor recruitment progress, receive AI-generated insights, and track hiring performance through interactive dashboards and reports.

By utilizing Salesforce technologies such as **Agentforce**, **Prompt Builder**, **Flow Automation**, **Approval Processes**, and **Reports & Dashboards**, the system significantly reduces manual effort while improving recruitment accuracy, transparency, and operational efficiency.

Project Scope

The project focuses on automating the recruitment process from candidate application submission to final hiring approval.

The system includes:

- Candidate application management
- AI-driven candidate prioritization
- Intelligent salary recommendation
- Hiring risk analysis
- Automated approval workflows
- Email notifications and alerts
- Recruitment performance dashboards
- Secure role-based access
- AI-assisted recruiter support using Agentforce

The solution is designed for recruiters, hiring managers, and Salesforce administrators, providing a scalable and secure recruitment platform suitable for organizations of different sizes.

Key Features

The AI-Powered Job Recruitment System offers the following features:

- AI-based candidate prioritization
- Resume evaluation assistance
- Intelligent salary recommendations
- Hiring risk identification
- Automated recruitment workflows
- Salary approval automation
- Email alerts and notifications
- Centralized candidate management
- Agentforce-powered recruiter assistance
- Prompt Builder integration
- Real-time recruitment dashboards
- Secure access using Salesforce security features
- Recruitment analytics and reporting
- Reduced manual intervention
- Improved hiring efficiency and consistency

2. Business Objectives

The primary objective of the **AI-Powered Job Recruitment & Candidate Management System** is to transform the traditional recruitment process into an intelligent, automated, and efficient hiring platform using Salesforce and Artificial Intelligence.

The project aims to reduce manual effort by automating repetitive recruitment activities while enabling recruiters and hiring managers to make faster, data-driven decisions.

Objectives

2.1 AI-Assisted Recruitment

Implement Artificial Intelligence to analyze candidate information and provide intelligent recommendations that help recruiters identify the most suitable applicants for available job positions.

2.2 Candidate Prioritization

Automatically prioritize candidates based on predefined recruitment criteria such as skills, qualifications, experience, interview performance, and AI-generated insights. This enables recruiters to focus on the most promising applicants first.

2.3 Intelligent Salary Recommendations

Provide AI-generated salary recommendations based on candidate qualifications, experience level, and organizational hiring standards, ensuring consistency and fairness in compensation decisions.

2.4 Recruitment Workflow Automation

Automate recruitment activities including candidate application processing, interview scheduling, salary approvals, email notifications, and status updates using Salesforce Flows and Approval Processes.

2.5 Improve Decision Making

Support recruiters and hiring managers with AI-powered recommendations, hiring risk identification, and recruitment analytics to improve hiring accuracy and reduce subjective decision-making.

2.6 Role-Based Secure Access

Ensure that recruiters, hiring managers, and system administrators have secure access to only the information and functionalities relevant to their responsibilities through Salesforce security features.

2.7 Performance Monitoring

Provide real-time dashboards and reports that allow management to monitor recruitment progress, hiring efficiency, candidate status, approval timelines, and overall recruitment performance.

2.8 Scalable Recruitment Platform

Develop a flexible Salesforce solution that can easily accommodate future business requirements, additional AI capabilities, and increased recruitment volume without significant redesign.

3. Problem Statement

Recruitment is one of the most important business processes in any organization. However, many organizations still rely on manual recruitment methods, resulting in delays, inconsistent evaluations, communication gaps, and increased administrative effort.

Recruiters often spend significant time reviewing resumes, comparing candidates, coordinating interviews, obtaining approvals, and communicating with applicants.

Manual processes also increase the possibility of human errors, inconsistent hiring decisions, and delayed recruitment cycles.

Furthermore, organizations frequently lack centralized visibility into recruitment performance, making it difficult to identify hiring bottlenecks, monitor recruiter productivity, and measure overall recruitment effectiveness.

To address these challenges, an AI-powered recruitment platform is required that combines automation, intelligent decision support, and centralized candidate management within a secure Salesforce environment.

The proposed system integrates Artificial Intelligence, Salesforce Flows, Agentforce, Prompt Builder, and Approval Processes to streamline hiring operations while improving efficiency, transparency, and candidate experience.

4. Existing System

Before implementing the proposed solution, the recruitment process was primarily manual and dependent on recruiters performing multiple repetitive tasks.

Limitations of the Existing System

- Manual collection and management of candidate information.
- Time-consuming resume screening and candidate evaluation.
- Subjective hiring decisions based on individual judgment.
- Delays in salary approvals and recruitment workflows.
- Limited visibility into recruitment performance.
- Manual email communication with candidates.
- Difficulty in tracking candidate application status.
- Lack of centralized recruitment data.
- No AI support for recruitment decisions.
- Increased administrative workload for recruiters.
- Higher possibility of human errors.
- Limited reporting and analytical capabilities.

These limitations reduced recruitment efficiency, increased hiring time, and made it difficult to ensure consistent recruitment decisions.

5. Proposed System

The **AI-Powered Job Recruitment & Candidate Management System** addresses the limitations of traditional recruitment by leveraging Salesforce automation and Artificial Intelligence.

The solution centralizes the complete recruitment lifecycle on the Salesforce platform, enabling recruiters and hiring managers to manage candidate applications, receive AI-generated recommendations, automate workflows, and monitor recruitment performance through dashboards and reports.

The system integrates Salesforce technologies such as **Agentforce, Prompt Builder, Salesforce Flow, Approval Processes, Reports, and Dashboards** to create an intelligent recruitment environment.

Key Functionalities

- Centralized candidate application management.
- AI-powered candidate prioritization.
- Intelligent salary recommendations.
- Hiring risk identification.
- Automated interview and recruitment workflows.
- Salary approval automation.
- Email notifications and alerts.
- Secure role-based access control.
- Agentforce assistance for recruiters.
- Real-time recruitment reports and dashboards.
- Automated status tracking throughout the hiring process.
- Reduced manual effort and improved operational efficiency.

Advantages of the Proposed System

- Faster recruitment process.
- Improved hiring accuracy.
- Consistent and data-driven decision making.
- Reduced manual workload.
- Enhanced recruiter productivity.
- Better candidate experience.
- Increased transparency in hiring.

- Secure management of recruitment data.
- Scalable architecture for future enhancements.
- Efficient monitoring through analytics and dashboards.

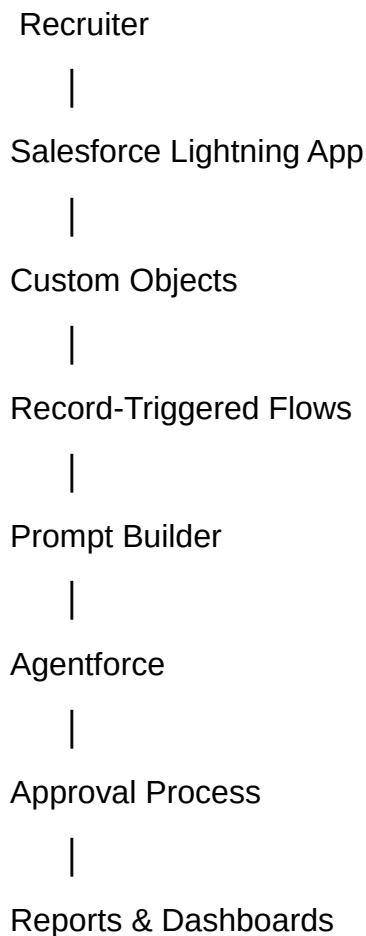
6. System Architecture

Overview

The AI-Powered Recruitment & Candidate Management System is built on the Salesforce platform and combines Artificial Intelligence with automation to simplify and accelerate the recruitment process. The architecture integrates Salesforce standard features, custom objects, Flow Automation, Agentforce, Prompt Builder, Approval Processes, Reports, and Dashboards into a centralized solution.

The system allows recruiters to manage candidate applications efficiently while AI assists in prioritizing candidates, generating hiring insights, and recommending salary packages. Automated workflows reduce manual intervention and ensure a smooth recruitment lifecycle.

Architecture Diagram



Architecture Components

User Layer

The system is accessed by different users based on their roles:

- Recruiter
- Hiring Manager
- Salesforce Administrator

Each user has role-based permissions to access only the required data and functionalities.

Application Layer

The application layer consists of Salesforce Lightning components and business logic, including:

- Custom Objects
- Lightning Record Pages
- Screen Flows
- Approval Processes
- Validation Rules
- Email Alerts
- Reports and Dashboards

This layer controls the recruitment workflow and business rules.

AI Layer

Artificial Intelligence capabilities are provided through Salesforce Agentforce and Prompt Builder.

The AI layer performs:

- Candidate prioritization
- Hiring recommendations
- Salary suggestions
- Risk identification
- Recruiter assistance through conversational AI

These AI-generated insights help recruiters make faster and more informed hiring decisions.

Data Layer

All recruitment information is securely stored within Salesforce.

The data layer includes:

- Candidate records
- Job positions
- Applications
- Interview details
- Salary approvals
- Recruitment reports

Salesforce maintains data integrity through validation rules, relationships, and security settings.

7. Data Model

Overview

The recruitment system follows a structured data model that stores candidate information and recruitment activities in related Salesforce objects.

The data model ensures accurate data management, efficient reporting, and seamless automation throughout the recruitment process.

Main Objects

Candidate

Stores candidate information such as:

- Candidate Name
- Email Address
- Phone Number
- Education
- Experience
- Skills
- Resume
- Current Status

Job Position

Stores job-related information including:

- Job Title

- Department
- Location
- Required Skills
- Salary Range
- Hiring Manager

Candidate Application

Acts as the central object connecting candidates with job positions.

Stores:

- Application Date
- Application Status
- AI Priority Score
- Interview Status
- Recruiter
- Final Decision

Interview

Maintains interview details including:

- Interview Date
- Interview Type
- Interviewer
- Feedback
- Interview Result

Salary Approval

Stores approval information including:

- Recommended Salary
- Approved Salary
- Approval Status
- Approver
- Approval Date

Object Relationships

The objects are connected as follows:

- One Candidate can submit multiple Candidate Applications.
- One Job Position can receive multiple Candidate Applications.
- One Candidate Application can have one or more Interviews.
- Each Candidate Application can generate one Salary Approval request.
- Reports retrieve data from these related objects to provide recruitment analytics.

The structured relationships eliminate duplicate data and ensure consistency across the application.

8. AI-Powered Recruitment Design

Artificial Intelligence is one of the core features of this recruitment system. Instead of relying entirely on manual evaluation, the system provides intelligent recommendations that assist recruiters during the hiring process.

The AI components analyze candidate information and generate meaningful insights that improve recruitment accuracy and efficiency.

AI Features

Candidate Prioritization

The system evaluates candidate information and assigns a priority level based on predefined recruitment criteria.

This enables recruiters to focus first on the most suitable candidates.

Hiring Recommendations

AI assists recruiters by generating hiring recommendations based on candidate qualifications, skills, and application details.

These recommendations act as decision-support rather than replacing human judgment.

Salary Recommendation

Based on candidate profile information, AI suggests an appropriate salary range that aligns with organizational hiring standards.

This improves fairness and consistency in compensation decisions.

Hiring Risk Identification

The system identifies potential recruitment risks such as missing qualifications, incomplete information, or other factors that may require additional review before hiring decisions are made.

Recruiter Assistance

Recruiters can interact with Agentforce to quickly obtain AI-generated summaries, recommendations, and guidance during recruitment activities.

9. Agentforce Configuration

Agentforce is integrated into the recruitment system to provide conversational AI assistance for recruiters and hiring managers.

Instead of manually analyzing every application, recruiters can use Agentforce to obtain intelligent insights and recommendations directly within Salesforce.

Agentforce Capabilities

- Provides AI-generated candidate summaries.
- Assists recruiters during candidate evaluation.
- Generates hiring recommendations.
- Supports salary recommendation decisions.
- Identifies recruitment risks.
- Reduces manual analysis effort.
- Improves recruitment productivity.
- Enhances decision-making consistency.

Benefits of Agentforce

- Faster candidate evaluation.
- Improved hiring accuracy.
- Reduced recruiter workload.
- Better user experience.
- Increased operational efficiency.
- Intelligent recruitment support within Salesforce.

The integration of Agentforce makes the recruitment process more efficient by combining AI intelligence with Salesforce automation while keeping recruiters in control of final hiring decisions.

10. Prompt Builder Configuration

Overview

Prompt Builder is used to generate AI-powered responses within the recruitment system. It enables recruiters to receive intelligent recommendations based on candidate information stored in Salesforce.

The prompts are designed to analyze candidate details and produce meaningful outputs that support recruitment decisions while maintaining consistency across the hiring process.

Prompt Builder Objectives

The Prompt Builder implementation aims to:

- Generate AI-based candidate summaries.
- Recommend suitable candidates for job positions.
- Provide salary recommendations based on candidate profiles.
- Identify potential hiring risks.
- Assist recruiters in making informed hiring decisions.
- Reduce manual evaluation time.

Prompt Design

The prompts are configured to analyze candidate information such as:

- Educational qualification
- Technical skills
- Work experience
- Interview performance
- Application status
- Job requirements

Based on this information, AI generates recommendations that assist recruiters during candidate evaluation.

Benefits of Prompt Builder

- Faster candidate analysis.
- Consistent AI-generated recommendations.
- Improved recruiter productivity.
- Reduced manual effort.

- Better hiring accuracy.
 - Enhanced decision-making support.
-

11. Salesforce Flow Automation

Overview

Salesforce Flow is the primary automation tool used in this project. It automates repetitive recruitment activities and ensures that business processes are executed consistently without manual intervention.

The implemented flows guide recruiters through candidate management, application processing, approval routing, and notification handling.

Flow Objectives

The automation flows are designed to:

- Reduce manual work.
- Improve recruitment efficiency.
- Standardize business processes.
- Minimize human errors.
- Automatically update records.
- Trigger notifications and approvals.

Flow 1 – Candidate Application Creation

A Screen Flow provides recruiters with a guided interface to create new candidate applications.

Process

1. Recruiter opens the Candidate Application Screen Flow.
2. Candidate information is entered.
3. Job Position is selected.
4. Required details are validated.
5. Candidate Application record is created.
6. Confirmation message is displayed.

Benefits

- User-friendly interface.

- Reduced data entry errors.
 - Standardized application process.
 - Faster record creation.
-

Flow 2 – Candidate Status Update

This flow automatically updates candidate status as recruitment progresses.

Possible status values include:

- Applied
- Under Review
- Interview Scheduled
- Interview Completed
- Selected
- Rejected
- Offer Released

The flow ensures that recruiters always have accurate recruitment progress information.

Flow 3 – AI Recommendation Trigger

Whenever candidate information is updated, the system triggers AI evaluation.

The AI generates:

- Candidate priority
- Hiring recommendation
- Salary recommendation
- Risk indicators

These recommendations become available for recruiters to review before making hiring decisions.

Flow 4 – Notification Flow

Automated notifications are sent whenever important recruitment events occur.

Examples include:

- Candidate application submitted.
- Interview scheduled.
- Salary approval requested.
- Candidate selected.
- Candidate rejected.
- Recruitment process completed.

This keeps recruiters and hiring managers informed throughout the recruitment lifecycle.

12. Approval Process

Overview

The Approval Process ensures that salary decisions and final hiring approvals follow organizational policies before being finalized.

Instead of allowing direct approvals, Salesforce automatically routes approval requests to the appropriate hiring manager.

Approval Workflow

Step 1

Recruiter evaluates the candidate.



Step 2

AI generates hiring and salary recommendations.



Step 3

Recruiter submits the approval request.



Step 4

Hiring Manager reviews the request.



Step 5

Manager approves or rejects the recommendation.



Step 6

Candidate status is automatically updated.



Step 7

Email notifications are sent to relevant users.

Benefits

- Standardized approval process.
 - Improved transparency.
 - Faster decision making.
 - Better compliance with recruitment policies.
 - Complete approval history for auditing purposes.
-

13. Email Templates and Notifications

Overview

The recruitment system uses Salesforce Email Templates and automated Email Alerts to improve communication with recruiters, hiring managers, and candidates.

These notifications are generated automatically through Salesforce Flow and Approval Processes.

Email Notifications

The system sends notifications for various recruitment events, including:

- Candidate application confirmation.
- Interview scheduling.
- Interview reminders.
- Salary approval request.
- Approval completion.
- Candidate selection.
- Candidate rejection.
- Recruitment completion.

Advantages

- Faster communication.
- Reduced manual follow-up.
- Improved candidate experience.
- Consistent messaging.
- Better recruitment coordination.
- Increased operational efficiency.

The integration of Email Templates with Salesforce automation ensures timely communication throughout every stage of the recruitment process while reducing administrative workload.

14. Security Configuration

Overview

Security is an essential component of the AI-Powered Recruitment & Candidate Management System. Salesforce provides multiple layers of security to ensure that recruitment data is protected and only accessible to authorized users.

The system implements role-based access control, user authentication, object-level permissions, and field-level security to maintain confidentiality, integrity, and compliance.

Security Components

Profiles

Profiles define the permissions assigned to different users.

The project includes the following profiles:

- System Administrator
- Recruiter
- Hiring Manager

Each profile has access only to the features required for its responsibilities.

Roles

Salesforce Roles are used to manage record-level access.

Example role hierarchy:

- System Administrator
- Hiring Manager
- Recruiter

This hierarchy allows managers to monitor recruitment activities while ensuring recruiters can manage only their assigned records.

Permission Sets

Permission Sets provide additional permissions without modifying user profiles.

They are used to grant access to specific Salesforce features such as:

- Agentforce
- Prompt Builder
- Reports
- Dashboards

Object-Level Security

Object permissions control Create, Read, Edit, and Delete (CRUD) operations for recruitment-related objects.

Objects secured include:

- Candidate
- Job Position
- Candidate Application
- Interview
- Salary Approval

Field-Level Security

Sensitive fields such as salary recommendations, approval decisions, and recruiter comments are protected using Field-Level Security to prevent unauthorized access.

Benefits

- Secure recruitment data.
- Role-based access control.
- Protection of confidential candidate information.
- Compliance with organizational security policies.
- Reduced risk of unauthorized data modification.

15. Reports and Dashboards

Overview

Reports and Dashboards provide real-time visibility into recruitment activities, enabling recruiters and managers to monitor hiring performance and make informed decisions.

Reports

The system includes reports such as:

- Candidate Applications Report
- Recruitment Status Report
- Interview Schedule Report
- Salary Approval Report
- Recruiter Performance Report
- Job Position Report
- Candidate Selection Report

These reports help track recruitment progress and identify bottlenecks.

Dashboards

Interactive dashboards display key recruitment metrics including:

- Total Applications
- Candidates Under Review
- Interviews Scheduled
- Selected Candidates
- Rejected Candidates
- Pending Salary Approvals
- Recruitment Performance Trends

Benefits

- Real-time recruitment monitoring.
- Improved decision making.
- Better workforce planning.
- Increased operational transparency.
- Performance measurement through visual analytics.

16. Testing and Validation

Overview

Comprehensive testing was conducted to ensure that all Salesforce components function correctly and meet the project requirements.

Testing Activities

Functional Testing

Verified that:

- Candidate records are created successfully.
- Screen Flows execute correctly.
- Approval Processes function as expected.
- Email notifications are triggered.
- Reports display accurate data.

AI Validation

Confirmed that Agentforce and Prompt Builder generate relevant hiring recommendations and salary suggestions based on candidate information.

Security Testing

Verified that users can access only the records and features permitted by their assigned profiles and roles.

User Acceptance Testing (UAT)

The complete recruitment workflow was tested by simulating recruiter and hiring manager activities to ensure the solution met business requirements.

Testing Outcome

The system successfully passed all major functional and security tests, demonstrating reliable performance and readiness for deployment.

17. Deployment Readiness

Before deployment, the following activities were completed:

- Validation of Salesforce Flows.
- Verification of Approval Processes.
- Configuration of Agentforce and Prompt Builder.
- Testing of Reports and Dashboards.
- Security configuration review.

- Email notification verification.
- End-to-end recruitment workflow testing.
- User Acceptance Testing.

The project is deployment-ready and can be implemented in a Salesforce production environment.

18. Monitoring and Maintenance

After deployment, continuous monitoring is necessary to ensure system reliability and optimal performance.

Monitoring Activities

- Review Flow execution logs.
- Monitor Approval Process performance.
- Analyze Agentforce responses.
- Verify Prompt Builder outputs.
- Track email delivery status.
- Monitor dashboard accuracy.
- Review user feedback.
- Audit security settings periodically.

Maintenance Activities

- Update Flows when business processes change.
 - Modify Prompt Templates as recruitment requirements evolve.
 - Add new reports and dashboards when required.
 - Update user permissions.
 - Perform regular Salesforce maintenance.
 - Review system performance periodically.
-

19. Future Enhancements

The system has been designed with scalability in mind and can be enhanced with additional AI and Salesforce capabilities.

Possible future improvements include:

- Resume parsing using advanced AI.
- Automated interview scheduling with calendar integration.
- Candidate skill matching using machine learning.
- AI-powered interview question generation.
- Mobile recruiter application.
- Integration with LinkedIn and external job portals.
- Predictive hiring analytics.
- Employee onboarding automation.
- Multi-language recruitment support.
- Enhanced AI insights using future Agentforce capabilities.

These enhancements will further improve recruitment efficiency and user experience.

20. Conclusion

The AI-Powered Job Recruitment & Candidate Management System successfully demonstrates how Salesforce CRM, Flow Automation, Prompt Builder, and Agentforce can modernize the recruitment process. The project reduces manual effort, improves hiring efficiency, automates workflows, and provides intelligent AI-powered recommendations. Developed by Lakshmi Aiswarya Akella and Radhika Satya Malleswari Mandapati as part of the SmartBridge Internship, the solution showcases practical implementation of Salesforce technologies in solving real-world recruitment challenges.